Machine Learning and Computational Statistics
2nd Homework

Ex. 1

Consider the following non-linear model:

$$y = 3x_1^2 + 4x_2^2 + 5x_3^2 + 7x_1x_2 + x_1x_3 + 9x_2x_3 - 2x_1 - 3x_2 - 5x_3 + 4$$

Define a suitable function φ that transforms the problem to a space where the problem of estimating the model becomes linear.
What is the dim of the original and the transformed spaces?

$$\varphi(x) = [x_1^2 \; x_2^2 \; x_3^2 \; x_1x_2 \; x_1x_3 \; x_2x_3 \; x_1 \; x_2 \; x_3]^T$$

$(x = [x_1 \; x_2 \; x_3]^T)$

Original dim: 3
Transformed dim: 9

Ex. 2

Consider the following 2-class non-linear classification task:

$$x = [x_1 \; x_2 \; x_3]^T : x_1^2 + 3x_2^2 + 6x_3^2 + x_1x_2 + x_2x_3 \gtrless 3 \rightarrow x \in w_1 (w_2)$$

Define a suitable function φ that transforms the problem to a space where the problem of estimating the border of the 2 classes becomes linear.
What is the dim of the original and the transformed spaces?

$$\varphi(x) = [x_1^2 \; x_2^2 \; x_3^2 \; x_1x_2 \; x_2x_3]^T$$

Original dim: 3
Transformed dim: 5

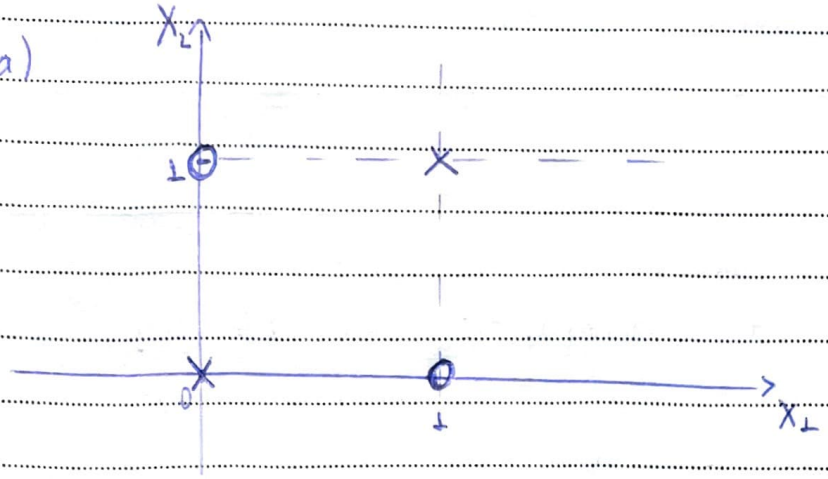
Ex. 3

Consider the exclusive OR (XOR) classification problem, where the points (0,0) and (1,1) are assigned to class −1 and the points (0,1) and (1,0) are assigned to class +1.

(a) Draw the points on the paper and prove that the classification problem is not linearly separable.

(b) Propose a transformation $\varphi(\cdot)$ that maps the above points to a new space where the XOR classification problem becomes linearly separable.

(a)



Let the classification problem be linearly separable. $\Rightarrow$
There exists a hyperplane (a line in 2D) defined by the equation $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$ that separates the 2 classes.

$(0,0)$ ∈ class -1 : $\theta_0 + \theta_1 \cdot 0 + \theta_2 \cdot 0 < 0 \Rightarrow \theta_0 < 0$ ①

$(1,1)$ ∈ class -1 : $\theta_0 + \theta_1 \cdot 1 + \theta_2 \cdot 1 < 0 \Rightarrow \theta_0 + \theta_1 + \theta_2 < 0$ ②

$(0,1)$ ∈ class $+1$: $\theta_0 + \theta_1 \cdot 0 + \theta_2 \cdot 1 > 0 \Rightarrow \theta_0 + \theta_2 > 0$ ③

$(1,0)$ ∈ class $+1$: $\theta_0 + \theta_1 \cdot 1 + \theta_2 \cdot 0 > 0 \Rightarrow \theta_0 + \theta_1 > 0$ ④

③ + ④ $\Rightarrow$ $2\theta_0 + \theta_1 + \theta_2 > 0 \overset{②}{\Rightarrow} \theta_0 < 0$ contradiction (①)

$\Rightarrow$ Our initial hypothesis is wrong $\Rightarrow$ The problem is n't linearly separable

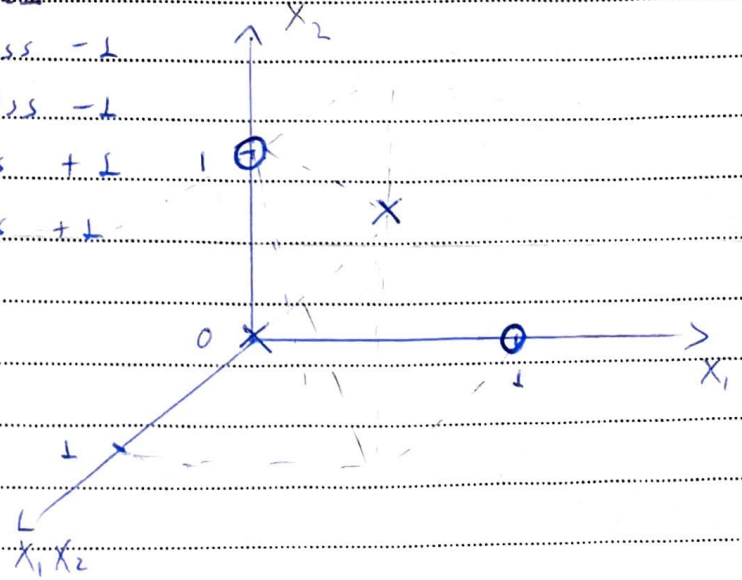
(b) $\varphi(x) = [x_1 \quad x_2 \quad x_1 x_2]^T$

$(0,0) \rightarrow (0,0,0)$ for class -1

$(1,1) \rightarrow (1,1,1)$ for class -1

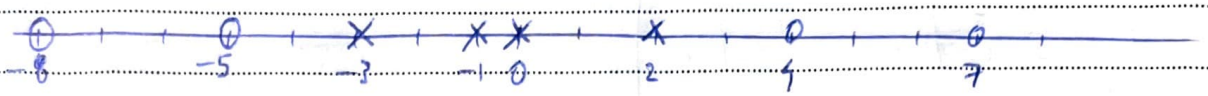
$(1,0) \rightarrow (1,0,0)$ for class $+1$

$(0,1) \rightarrow (0,1,0)$ for class $+1$



Ex. 4

Consider a 2-class one-dim classification problem where the points $-3, -1, 0, 2$ belong to class -1 and the points $-8, -5, 4, 7$ belong to class $+1$. Propose a transformation $g(\cdot)$ that maps the above points to a new one-dim space, where the classification problem becomes linearly separable.



$g(x) = x^2$

$-8 \to 64$
$-5 \to 25$
$4 \to 16$
$7 \to 49$

$-3 \to 9$
$-1 \to 1$
$0 \to 0$
$2 \to 4$

Observation: The points (original) of class -1 are clustered near the origin, possessing small absolute values.
The points (original) of class $+1$ are distributed at the 2 extremes of the axis, possessing larger absolute values.
The transformation I chose folds the axis, pushing the inner cluster to the left and the outer cluster to the right in the new space, thereby achieving linear separability.

Ex. 5

Consider a 2-class, 2-dim classification problem, where the data points $[1,1]^T$ $[1,2]^T$, $[2,1]^T$ belong to class $+1$, while the data points $[-1,-1]^T$, $[-1,-2]^T$, $[-2,-1]^T$ belong to class -1.

(a) Determine the hyperplane (line) that separates the data from the 2 classes, utilizing the LS criterion.

(b) Draw the data points from the classes along with the line determined in (a).

(c) Is this the unique line that separates the data from the 2 classes? Is this line the only one that results from the minimization of the sum of squared error criterion?

(a) Let the data points for class +1 ($y=1$) be:

$$x_1 = [1, 1]^T \quad x_2 = [1, 2]^T \quad x_3 = [2, 1]^T$$

Let the data points for class −1 ($y=-1$) be:

$$x_4 = [-1, -1]^T \quad x_5 = [-1, -2]^T \quad x_6 = [-2, -1]^T$$

The linear model: $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -2 \\ 1 & -2 & -1 \end{bmatrix} \qquad y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

The LS estimate is given by $\hat{\theta} = (X^T X)^{-1} X^T y$

$$\rightsquigarrow X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & -1 & -1 & -2 \\ 1 & 2 & 1 & -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -2 \\ 1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 12 & 10 \\ 0 & 10 & 12 \end{bmatrix}$$

$\rightsquigarrow (X^T X)^{-1}$ computation

$$\left(\begin{array}{ccc|ccc} 6 & 0 & 0 & 1 & 0 & 0 \\ 0 & 12 & 10 & 0 & 1 & 0 \\ 0 & 10 & 12 & 0 & 0 & 1 \end{array}\right) \xrightarrow[r_3 = r_3 - g_{32} r_2]{g_{32} = \frac{10}{12} = \frac{5}{6}} \left(\begin{array}{ccc|ccc} 6 & 0 & 0 & 1 & 0 & 0 \\ 0 & 12 & 10 & 0 & 1 & 0 \\ 0 & 0 & 11/3 & 0 & -5/6 & 1 \end{array}\right)$$

$$\xrightarrow[r_2 = r_2 - g_{23} r_3]{g_{23} = \frac{10}{11/3} = \frac{30}{11}} \left(\begin{array}{ccc|ccc} 6 & 0 & 0 & 1 & 0 & 0 \\ 0 & 12 & 0 & 0 & 36/11 & -30/11 \\ 0 & 0 & 11/3 & 0 & -5/6 & 1 \end{array}\right) \xrightarrow[r_3 = \frac{3}{11} r_3]{\substack{r_1 = \frac{1}{6} r_1 \\ r_2 = \frac{1}{12} r_2}}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/6 & 0 & 0 \\ 0 & 1 & 0 & 0 & 36/132 & -30/132 \\ 0 & 0 & 1 & 0 & -15/66 & 3/11 \end{array}\right)$$

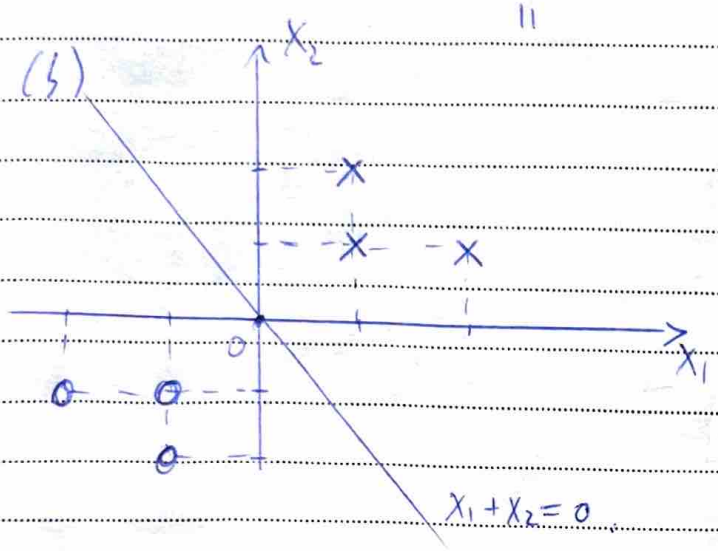
$$\Rightarrow (X^T X)^{-1} = \begin{pmatrix} 1/6 & 0 & 0 \\ 0 & 3/11 & -15/66 \\ 0 & -15/66 & 3/11 \end{pmatrix} = \begin{pmatrix} 1/6 & 0 & 0 \\ 0 & 3/11 & -5/22 \\ 0 & -5/22 & 3/11 \end{pmatrix}$$

$$\rightsquigarrow X^T y = \begin{bmatrix} 0 \\ 6 \\ 6 \end{bmatrix}$$

$$\rightsquigarrow \hat{\theta} = (X^T X)^{-1} (X^T y) = \begin{bmatrix} 0 \\ 4/11 \\ 4/11 \end{bmatrix}$$

$$f(x) = 0 \Rightarrow 0 + \frac{4}{11} x_1 + \frac{4}{11} x_2 = 0 \Rightarrow \boxed{x_1 + x_2 = 0}$$

(5)



(c) There are infinitely many lines that can separate these 2 perfectly separable classes. Although, this is the only line that results from the LS criterion, because the equation $(X^T X)\hat{\theta} = X^T y$ gives a unique solution for $\hat{\theta}$.

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